

Negative Exponents with Variable Bases

MA.8.AR.1.1

I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

Yesterday every exponent on a letter was positive.

Today x^3 arrives — and it is not a negative number.

Warm-Up

On your own. Four minutes. No notes.

1. Write as a single monomial: $(3x^2)^3$
2. Find the value: 5^{-2}
3. Which is greater, $\sqrt{80}$ or 9?
4. What is the reciprocal of $\frac{2}{3}$?

Warm-Up

Answers.

1. Write as a single monomial: $(3x^2)^3$ **$27x^6$**

2. Find the value: 5^{-2} **$1/25$**

3. Which is greater, $\sqrt{80}$ or 9? **9**

4. What is the reciprocal of $\frac{2}{3}$? **$\frac{3}{2}$**

Notes

Copy the table.

I. The pattern, with a letter

$\mathbf{x^4}$	$\mathbf{x^3}$	$\mathbf{x^2}$	$\mathbf{x^1}$	$\mathbf{x^0}$	$\mathbf{x^{-1}}$	$\mathbf{x^{-2}}$	$\mathbf{x^{-3}}$
$\mathbf{x \cdot x \cdot x \cdot x}$	$\mathbf{x \cdot x \cdot x}$	$\mathbf{x \cdot x}$	$\mathbf{x}$	$\mathbf{1}$	$\mathbf{1/x}$	$\mathbf{1/x^2}$	$\mathbf{1/x^3}$

$$x^{-n} = \frac{1}{x^n}$$

$$\frac{1}{x^{-n}} = x^n$$

$$(\quad x \neq 0 \quad)$$

Notes

II. Moving factors across the fraction bar

$$3x^{-2} = \frac{3}{x^2}$$

$$y^{-3}z^{-4} = \frac{1}{y^3z^4}$$

$$\frac{a^2}{b^{-3}} = a^2b^3$$

$$\frac{5x^{-4}y^2}{z^{-1}} = \frac{5y^2z}{x^4}$$

Only the factor with the negative exponent moves. Its exponent turns positive.

Notes

III. The laws with negative exponents

$$x^4 \cdot x^{-6} = x^{4+(-6)} = x^{-2} = \frac{1}{x^2}$$

$$\frac{m^3}{m^7} = m^{3-7} = m^{-4} = \frac{1}{m^4}$$

$$(2y^{-3})^2 = 2^2 \cdot y^{-6} = \frac{4}{y^6}$$

Apply the law. Then rewrite so that every exponent is positive.

Example 1

Multiplying monomials with negative exponents.

$$(2x^{-3}y^2)(6x^5y^{-4})$$

Write an equivalent expression with positive exponents only.

Example 1

Worked.

$$(2x^{-3}y^2)(6x^5y^{-4}) = 12 \cdot x^{-3+5} \cdot y^{2+(-4)} \quad \text{multiply coefficients; add exponents}$$
$$= 12x^2y^{-2} = \frac{12x^2}{y^2} \quad \text{move } y^{-2} \text{ down; the 12 and the } x^2 \text{ stay up}$$

Answer: $12x^2/y^2$

Your Turn

Same steps, your numbers. Boards up when done.

$$(4a^{-2}b)(a^5b^{-3})$$

Your Turn

Answer.

$$(4a^{-2}b)(a^5b^{-3})$$

$$4 \cdot a^3 \cdot b^{-2}$$

$$\textbf{Answer: } \frac{4a^3}{b^2}$$

Example 2

Several laws in one expression — the state guide's item.

$$\left(\frac{a^3}{4b^{-7}}\right)^5$$

Write an equivalent expression with positive exponents only.

Example 2

Worked.

$$\left(\frac{a^3}{4b^{-7}}\right)^5 = \frac{a^{15}}{4^5 b^{-35}} \quad \text{raise every factor: } a^3, 4, b^{-7}$$

$$= \frac{a^{15} b^{35}}{1,024} \quad 4^5 = 1,024; b^{-35} \text{ moves up as } b^{35}$$

Answer: $a^{15}b^{35}/1,024$

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{x^2}{3y^{-2}}\right)^3$$

Your Turn

Answer.

$$\left(\frac{x^2}{3y^{-2}}\right)^3$$

x^6 over $27y^{-6}$, then move the y up

Answer: $\frac{x^6y^6}{27}$

Whiteboards • Question 1 of 9

Boards up on three.

$$x^{-4}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 1 of 9

Answer.

$$x^{-4}$$

reciprocal of x^4

Answer: $\frac{1}{x^4}$

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{1}{y^{-5}}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 2 of 9

Answer.

$$\frac{1}{y^{-5}}$$

the power moves up

Answer: y^5

Whiteboards • Question 3 of 9

Boards up on three.

$$3x^{-2}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 3 of 9

Answer.

$$3x^{-2}$$

only the x moves

Answer: $\frac{3}{x^2}$

Whiteboards • Question 4 of 9

Boards up on three.

$$y^{-3}z^{-4}$$

Answer it.

Rewrite with positive exponents.

Whiteboards • Question 4 of 9

Answer.

$$y^{-3}z^{-4}$$

both factors move down

Answer: $\frac{1}{y^3z^4}$

Whiteboards • Question 5 of 9

Boards up on three.

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

A. $1/x^2$

B. x^{10}

C. $1/x^{24}$

D. $-x^2$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

A. $1/x^2$

B. x^{10}

C. $1/x^{24}$

D. $-x^2$

Answer: A — $1/x^2$

Whiteboards • Question 6 of 9

Boards up on three.

$$(2x^{-3})^2$$

Which expression is equivalent?

A. $4/x^6$

B. $2/x^6$

C. $4/x^5$

D. $1/(4x^6)$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2x^{-3})^2$$

Which expression is equivalent?

A. $4/x^6$

B. $2/x^6$

C. $4/x^5$

D. $1/(4x^6)$

Answer: A — $4/x^6$

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{m^3}{m^7}$$

Answer it.

Write with a positive exponent.

Whiteboards • Question 7 of 9

Answer.

$$\frac{m^3}{m^7}$$

m^{-4} — then rewrite

Answer: $\frac{1}{m^4}$

Whiteboards • Question 8 of 9

Boards up on three.

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

Answer it.

Which law applies?

Whiteboards • Question 8 of 9

Answer.

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

$$-5 + n = 2$$

Answer: $n = 7$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

Write your answer in sentences.

Use the word reciprocal.

Whiteboards · Question 9 of 9

Answer.

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

reciprocals: 8 and $1/8$

Answer: x^{-3} is the reciprocal of x^3 . When $x = 2$: $x^3 = 8$ and $x^{-3} = 1/8$.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Variables with negative exponents — 2NL

Multiply and divide powers: variable bases — 85P